DRAGOON MOUNTAINS GROWING GUIDES

Growing Peppers in the Dragoon Mountains

A Complete Guide to All Chile Varieties

Dragoon Mountains, Cochise County, Arizona

Home of the Native Chiltepin

Contents

Figure 1 – The Dragoon Mountains rising above the Sulphur Springs Valley, Cochise County, Arizona.

Chiltepin · Jalapeño · Poblano · Cayenne · Serrano · Rocoto · Habanero · Ancho · Hatch · Bell & Sweet Varieties

Soil · Irrigation · Nutrition · Harvest · Overwintering

This guide is part of the Dragoon Mountains Growing Library:

- **Guide A — Growing Peppers in the Dragoon Mountains** (this guide) · soil peppers, all varieties
- **Guide B — Bato Bucket Hydroponic Setup** · complete pepper-system build
- **Guide C — Hydroponic Pepper Guide** · hydroponic pepper companion
- **Guide D — A Hydroponic Primer** · general-purpose hydroponic onramp
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1 · Why the Dragoons Are Perfect for Peppers

Native Chiltepin · climate heritage · your natural advantages

The Dragoon Mountains are not just suitable for pepper growing — they are ancestral pepper territory. The Chiltepin (*Capsicum annuum* var. *glabriusculum*), the wild ancestor of nearly all cultivated chiles, grows natively in Cochise County and the surrounding sky island ranges. The Apache people harvested and used it for centuries. You are growing peppers in their homeland.

Climate Advantages for Pepper Growing

Factor	Dragoon Conditions	Why Peppers Love It
Summer temperatures	85–95°F daytime	Ideal fruiting range. Hot enough to set fruit reliably without blossom drop from extreme heat.
Night temperatures	55–70°F — cool nights	Peppers set fruit best when nights drop below 75°F. Dragoon nights are perfect.
Monsoon humidity	50–70% July–Sept	Boosts fruit sizing and prevents blossom drop from dry air stress.
Monsoon rainfall	8–12” in 6 weeks	Reduces irrigation need by 40–60% during peak fruiting season.
Diurnal temperature swing	30–40°F daily	Concentrates capsaicin and flavour compounds. Dragoon chiles are hotter and more flavourful.
Fall season	Sept–Nov — warm days, cool nights	Extended ripening window. Peppers continue producing long after summer crops are done.
UV intensity	15–20% higher at elevation	Accelerates ripening and colour development. Intense sun deepens flavour.
Frost timing	First frost Oct–Nov at mid elevation	Long growing season — 180+ frost-free days at 4,500–5,000 ft.

The Chiltepin Connection

The wild Chiltepin grows along rocky canyon washes and under mesquite trees throughout Cochise County, including the Dragoon and Chiricahua mountain ranges. These small, intensely hot berries (50,000–100,000 SHU) disperse via birds and have survived here for thousands of years with zero irrigation and no amendments. They prove this land was made for peppers.

Wild Chiltepin plants can be found growing naturally along canyon washes in the Dragoons. Collect seeds from ripe berries in fall — these are perfectly adapted to your exact conditions and make excellent parent stock for your garden. They are also genuinely rare and delicious.

2 · Pepper Species & Variety Guide

Scoville ratings · days to maturity · Dragoon suitability

Peppers belong to five main domesticated species. Understanding which species each variety belongs to helps predict its temperature preferences, growing habit, and how well it will perform at Dragoon elevation.

Species Overview

Species & Key Facts	Dragoon Suitability
C. annuum · Jalapeño, poblano, cayenne, serrano, bell, Hatch, Anaheim, Chiltepin · *hot summers, warm nights*	★★★★★ **Excellent** — widest variety selection, most productive species here.
C. chinense · Habanero, Scotch bonnet, ghost pepper · *very hot — needs high heat*	★★★★ **Good** — lower Dragoon elevations best; may need shade cloth above 5,500 ft.
C. pubescens · Rocoto, Manzano · *cool highlands — prefers 45–65°F nights*	★★★★★ **Ideal** — the Dragoons' cool nights are exactly what Rocoto needs.
C. frutescens · Tabasco, Malagueta · *hot, humid*	★★★ **Moderate** — monsoon season works well; dry season challenging.
C. baccatum · Aji Amarillo, Lemon Drop · *warm, moderate*	★★★★ **Good** — Andean origin tolerates elevation and temperature swings.

Complete Variety Reference Guide

Variety · Profile	Dragoon Notes
Bell Pepper · *C. annuum* · 0 SHU · Sweet · 70–80d · 18–24”	Excellent — fall crop. Heat above 90°F causes blossom drop; plant for fall harvest.
Poblano / Ancho · *C. annuum* · 1,000–2,000 SHU · Mild · 65–80d · 24–30”	★ Excellent. Thick walls love the Dragoon fall. Green = poblano; dried red = ancho.
Anaheim / Hatch · *C. annuum* · 500–2,500 SHU · Mild · 75–80d · 24–30”	★ Excellent. New Mexico Hatch varieties thrive at similar elevations. Roast by the bushel.
Hungarian Wax · *C. annuum* · 1,000–15,000 SHU · Mild–Med · 70d · 18–24”	Good. Early ripening — useful for spring harvest before monsoon.
Jalapeño · *C. annuum* · 2,500–8,000 SHU · Medium · 70–75d · 24–30”	★ Excellent. Native to Mexico — loves the Dragoon climate. Hotter when stressed. Very productive.

Variety · Profile	Dragoon Notes
Serrano · *C. annuum* · 10,000–25,000 SHU · Hot · 75–90d · 18–24”	★ Excellent. More heat-tolerant than jalapeño. Prolific producer. Great for fresh salsa.
Chipotle (smoked jalapeño) · *C. annuum* · 2,500–8,000 SHU · Medium · 70–75d · 24–30”	Harvest jalapeños when red for traditional chipotle smoking over mesquite — perfect local wood.
Cayenne · *C. annuum* · 30,000–50,000 SHU · Hot · 70–75d · 18–24”	★ Excellent. Heat-tolerant, prolific, easy to dry at Dragoon elevation. Thin walls dry perfectly in the fall sun.
Chiltepin (native) · *C. annuum* · 50,000–100,000 SHU · Very Hot · 90–120d · 36–48”	★★ Native to Cochise County — zero adaptation needed. Perennial if protected from frost. Grows wild in canyon washes. The most important local variety.
Thai / Bird’s Eye · *C. annuum* · 50,000–100,000 SHU · Very Hot · 75–90d · 18–24”	Good. Compact plant, prolific, heat-tolerant. Excellent container variety.
Habanero · *C. chinense* · 100,000–350,000 SHU · Extreme · 85–100d · 24–36”	Good at lower elevations (< 5,000 ft). Needs long, hot season. Start 10–12 weeks early. Monsoon humidity improves fruit set.
Ghost / Bhut Jolokia · *C. chinense* · 800,000–1,000,000+ SHU · Extreme · 100–120d · 36–48”	Challenging — needs very long season. Start January indoors. Best below 4,800 ft elevation. Monsoon season critical for fruit set.
Scotch Bonnet · *C. chinense* · 100,000–350,000 SHU · Extreme · 90–100d · 24–36”	Moderate — similar to habanero. Fruity flavour worth the effort.
Rocoto (Red/Yellow) · *C. pubescens* · 30,000–100,000 SHU · Very Hot · 90–120d · 36–60”	★★ Ideal — the Dragoons’ cool nights are perfect for Rocoto. Perennial to 15+ years if protected from frost. See dedicated Chapter 3.
Manzano (Apple Chile) · *C. pubescens* · 12,000–30,000 SHU · Hot · 120+d · 36–60”	★★ Ideal — same as Rocoto. Yellow/orange apple-shaped fruit. Black seeds. Beautiful purple flowers.
Aji Amarillo · *C. baccatum* · 30,000–50,000 SHU · Hot · 90–100d · 36–48”	Good. Andean origin — elevation tolerant. Fruity, unique flavour. Excellent food pepper. Needs staking.
Lemon Drop · *C. baccatum* · 15,000–30,000 SHU · Hot · 90–100d · 24–36”	Good. Citrusy, bright flavour. Heat and drought tolerant. Excellent for salsas and hot sauce.

3 · The Rocoto Special Section

Capsicum pubescens · high-elevation adaptation · growing secrets

The Rocoto (*Capsicum pubescens*) is the rarest, most elevation-adapted, and most misunderstood pepper species in cultivation. Native to the Andes Mountains at 6,000–14,000 ft elevation, it is the only pepper species that thrives in cool conditions — and it is perfectly suited to the Dragoon Mountains. While most chiles struggle with cool Dragoon nights, the Rocoto absolutely requires them to set fruit.

Why Rocoto is Perfect for the Dragoons

Rocoto Requirement	Dragoon Condition	Match?
Cool nights 45–65°F for fruit set	55–68°F night temps most of the year	★★ Perfect
Partial shade tolerance in hot weather	Mountain aspect provides afternoon shade on east-facing slopes	★★ Perfect
Well-drained, deep soil	Sandy loam over caliche — break through hardpan at planting	✓ Good
pH 6.0–7.0	Dragoon soils pH 7.5–8.5 — needs sulfur amendment	■ Needs work
Moderate humidity during fruit set	Monsoon July–Sept provides 50–70% humidity	★★ Perfect
Protection from frost (perennial)	Frost arrives Oct–Nov — needs protection or harvest by then	✓ Manageable
Long growing season 90–120+ days	180+ frost-free days at mid elevation	✓ Sufficient

Rocoto Growing Protocol for the Dragoons

1. **Start very early — January indoors** — Rocotos need 10–14 weeks before transplant. Germination is slow (14–28 days). Use bottom heat mat at 75–80°F.
2. **Partial shade siting** — Plant on the east side of a structure or taller plants. Morning sun is fine — afternoon shade prevents blossom drop in summer heat.
3. **Amend soil heavily** — Rocoto needs pH 6.0–6.5. Apply sulfur in fall before planting. Add 4” compost. They are heavy feeders compared to other chiles.
4. **Consistent moisture** — Never let Rocotos dry out — this immediately causes blossom drop. Daily drip irrigation during dry periods. Mulch heavily.
5. **Patience at flowering** — Rocotos bloom prolifically but will drop flowers if nights exceed 75°F. The Dragoon monsoon season (cool nights returning) triggers the best

fruit set.

6. **Frost protection = perennial** — At the first frost threat in October, cover with frost cloth or bring container plants indoors. Protected Rocotos can live 15+ years and grow 10 ft tall — becoming a permanent garden feature.

Black seeds: Rocoto is the only domesticated pepper species with black seeds — a reliable identifier. It also has distinctly hairy leaves and stems, and beautiful purple-violet six-pointed star flowers. If your ‘rocoto’ has white flowers and white/tan seeds, it is mislabelled — check your seed source.

4 · Site Selection & Soil Preparation

Sun · drainage · pH 6.0–7.0 · amendments · raised beds

Peppers are sun-loving, warmth-requiring plants that reward good soil preparation generously. In the Dragoons, the main soil challenges are high alkalinity (pH 7.5–8.5) and low organic matter. Both are solvable with targeted amendments before planting.

Soil Requirements

Parameter	Target	Typical Dragoon Reading	Fix
pH	6.0 – 7.0	7.5 – 8.5	Elemental sulfur 1–2 lb per 100 sq ft. Acidify irrigation water to pH 6.5.
Organic matter	3 – 5%	< 1%	3–4” compost worked into top 10”. Repeat every season.
Drainage	Water infiltrates freely	Can waterlog over caliche	Raised beds 12–18” deep. Break caliche layer before planting.
Texture	Sandy loam — well-draining	Variable — often adequate	Add perlite or coarse sand to heavy clay areas. 20–30% by volume.
Nitrogen	Moderate pre-plant	Often low	Aged compost + alfalfa meal pre-plant. Avoid high-N during fruiting.
Calcium	Adequate — critical for BER prevention	High (abundant from caliche)	Usually sufficient. Issues arise from pH lockout or uneven watering, not deficiency.

Pre-Plant Bed Preparation

- ■ **Soil test first** — know your pH before adding anything. UA Cooperative Extension (520-458-8278) can advise on local testing.
- ■ **Apply elemental sulfur** — 1–2 lb per 100 sq ft if pH > 7.5. Work in 4–6 weeks before planting for best effect.

- ■ **Add 3–4” compost** — work into top 8–10” of soil. Local alfalfa hay compost from Willcox feed stores is excellent.
- ■ **Check caliche depth** — drive a rebar rod. If hardpan is shallower than 18”, build raised beds or auger through before planting.
- ■ **Raised beds** — the easiest solution for most Dragoon sites. 12–18” deep beds with a 50% native soil / 50% compost mix. Complete drainage control, optimal pH, no caliche limitation.

Companion Planting

Plant with:	Why
Basil	Repels aphids and thrips. Improves pepper flavour when grown nearby. Perfect Dragoon monsoon companion.
Carrots	Deep roots break up soil below pepper root zone. Don’t compete for same nutrients.
Oregano	Native-adapted, drought-tolerant. Repels pests and attracts beneficial insects. Perennial in Dragoons.
Marigolds	Strong repellent for nematodes and many insects. Plant around perimeter of pepper bed.
Onion / Garlic	Repel aphids and spider mites. Plant between pepper rows.

Avoid planting with:	Why
Tomatoes, potatoes, eggplant	Same solanaceae family — share diseases and pests. Rotate every 2–3 years.
Fennel	Allelopathic — inhibits pepper root development.
Brassicas (cabbage, broccoli)	Compete for same nutrients and may harbour aphids that spread to peppers.

5 · Starting Seeds & Transplanting

Indoor starts · heat mats · hardening off · timing

Peppers need a head start indoors — they are slow-growing and require warm soil for germination. At Dragoon elevations, the last frost occurs April–May, meaning seeds started in February give you the longest possible growing season.

Seed Starting Timeline by Variety

Variety Group	Start Indoors	Transplant Outdoors	Reason
Rocoto / Manzano (*C. pubescens*)	January	Late April – May	Slow germinators, very long season needed (120+ days)
Habanero, Ghost (*C. chinense*)	Late January – Feb	Late April – May	Longest season of common varieties (85–100 days)
Jalapeño, Serrano, Cayenne (*C. annuum*)	Mid February	Late April – early May	Standard 8–10 week head start
Poblano, Anaheim, Hatch, Bell	Mid February	Late April – early May	8–10 week head start, same as above
Chiltepin (native)	February – March	May	Scarify seeds first (light sandpaper) — tough seed coat
Aji, Lemon Drop (*C. baccatum*)	February	Late April – May	8–10 week head start

Germination Requirements

Factor	Target	Notes
Soil temperature	80–85°F (27–29°C)	The most critical factor. Use a heat mat under seed trays. Without heat, germination is slow and erratic.
Rocoto soil temp	75–80°F (24–27°C)	Rocoto prefers slightly cooler germination temps than other species.
Sowing depth	1/4” (6mm)	Cover lightly. Firm seed-to-soil contact essential.

Factor	Target	Notes
Germination time	7–21 days (*C. annuum*); 14–28 days (*C. pubescens*)	Rocoto and habanero are the slowest. Do not give up early.
Moisture	Consistently moist — not wet	Cover tray with clear dome until germination. Check daily.
Light (after sprout)	16 hrs under LED grow light	Place seedlings 2–3” below a T5 or quantum board LED immediately after sprouting.

Hardening Off & Transplanting

- ■ Begin hardening off **10–14 days before transplant date** — set plants outside in a sheltered spot for 1 hour on day 1, increasing by 1–2 hours daily.
- ■ **Never transplant when soil temperature is below 60°F** — check with a soil thermometer, not the calendar.
- ■ **Transplant on a cloudy day or in late afternoon** — reduces transplant shock from high-altitude UV.
- ■ **Space most varieties 18–24” apart** — Rocotos and large *C. chinense* need 30–36”.
- ■ **Water in immediately with a dilute fish emulsion solution** — 1 tbsp per gallon. This helps roots establish quickly.
- ■ **No nitrogen fertiliser for 2–3 weeks after transplant** — let roots establish before pushing growth.

6 · Irrigation & Water Management

Drip irrigation · monsoon management · blossom drop prevention

For hydroponic pepper production, see the *Bato Bucket Hydroponic Setup* guide (Guide B) and the *Hydroponic Pepper Guide* (Guide C). Those guides replace this chapter's drip-irrigation schedule with a reservoir-based watering plan.

Consistent moisture is more important for peppers than total water volume. Irregular watering — the classic Dragoon challenge with monsoon boom-and-bust cycles — causes blossom drop, blossom end rot, and split fruits. Drip irrigation plus thick mulch solves most of this automatically.

Irrigation by Growth Stage

Stage	Water Need	Frequency	Notes
Transplant establishment (Weeks 1–3)	Moderate — consistent	Daily light watering	Keep root zone moist. Don't overwater — no roots yet to absorb excess.
Vegetative growth (Weeks 3–8)	Moderate	Every 2 days	Increase as canopy grows. Drip at 0.5–1 GPH per plant.
Pre-flowering (Weeks 6–10)	Moderate — increase	Daily in dry heat	Consistent moisture critical as buds form. Drought stress now causes flower abortion.
Flowering (critical stage)	Consistent — never stress	Daily minimum	Blossom drop is triggered by both drought AND flooding. Keep even moisture.
Monsoon season (July–Sept)	Reduce drip	Monitor rainfall — reduce or pause drip	2" of rain = skip 3–5 days of drip. Mulch prevents oversaturation.
Fruiting (peak production)	Moderate–High	Daily or every 2 days	Large fruit load demands consistent water. Blossom end rot from irregular moisture.

Stage	Water Need	Frequency	Notes
Late harvest / wind-down	Reduce	Every 3 days	Slight water stress at this stage increases capsaicin concentration — hotter peppers.

Blossom Drop Prevention

Blossom drop is the most common frustration for Dragoon pepper growers. Understanding the triggers makes it preventable:

- ■ **Temperature:** Flowers drop when daytime temps exceed 90°F or nights drop below 55°F. The Dragoons rarely hit both extremes — but June pre-monsoon is the highest risk window.
- ■ **Water stress:** Even one day without water during flowering can drop an entire flush of blossoms. Install drip before plants flower.
- ■ **Overfertilising with nitrogen:** High-N pushes leaf growth instead of fruiting. Switch to low-N fertiliser at first flower bud.
- ■ **Thrips:** These tiny insects damage flowers directly. Yellow sticky traps near flowering plants catch them early.
- ■ **Monsoon salvation:** Many dropped buds are replaced within 2–3 weeks once monsoon rains bring cooler nights — the Dragoon monsoon often restores a previously struggling pepper plant.

7 · Nutrition & Fertilisation

NPK stages · calcium · magnesium · local sources

For hydroponic pepper production, see the *Bato Bucket Hydroponic Setup* guide (Guide B, Section 5) and the *Hydroponic Pepper Guide* (Guide C, Chapter 7). Those guides carry the Masterblend 3-part pepper programme with the 2.4 g/gal Epsom rule at fruit sizing.

Pepper nutrition changes significantly through the growing season. High nitrogen early builds plant structure; lower nitrogen with higher phosphorus and potassium at flowering and fruiting drives production. Calcium is the critical micromanagement nutrient — deficiency causes blossom end rot regardless of how well everything else is managed.

Seasonal Nutrition Programme

Stage	NPK Focus	Application	Local / Organic Source
Pre-plant soil prep	Balanced	3–4” compost + 5 lb alfalfa meal per 100 sq ft	Local alfalfa (Willcox), bat guano, aged manure from local ranches
Transplant establishment	Low N, root emphasis	Fish emulsion 1 tbsp/gal drench at planting	Fish emulsion (purchase), dilute compost tea
Vegetative growth	Moderate N, balanced P/K	Balanced fertiliser (10-10-10) every 2–3 weeks	Aerated bat guano tea; dilute fish emulsion + wood ash water
First flower buds visible	Reduce N, increase P/K	Switch to low-N formula (5-10-10 or similar)	Bone meal (P) + wood ash (K) + compost. Stop heavy nitrogen now.
Peak fruiting	Low N, high K and Ca	Potassium-rich liquid feed every 2 weeks	Sulfate of potash drench; kelp meal; caliche-vinegar calcium extract
Late season	Reduce all	Taper off — allow plant to focus on ripening	Foliar iron spray if yellowing appears from pH-related lockout

Calcium Management — Blossom End Rot Prevention

Blossom end rot (BER) — the dark sunken spot on the bottom of the fruit — is caused by calcium deficiency at the fruit, usually from irregular watering or pH issues that prevent calcium uptake, not from calcium absence in the soil. Dragoon soils have abundant calcium from caliche. The problem is getting it into the plant.

- ■ **Most important fix:** consistent, even watering via drip. BER disappears in most cases when moisture is stabilised.
- ■ **pH correction:** calcium locks out below pH 5.5 and above pH 8.0 — keeping pH at 6.0–7.0 keeps calcium available.
- ■ **Calcium foliar spray:** dissolve 1 tsp calcium nitrate per quart of water, spray directly on developing fruits in early morning.
- ■ **Avoid excess potassium:** high K competes with Ca uptake — don't overdo wood ash if BER is a problem.

Magnesium — Often Deficient in Sandy Dragoon Soils

- ■ **Symptom:** yellow leaves between green veins, starting on older leaves — classic Mg deficiency.
- ■ **Fix:** dissolve 1 tbsp Epsom salt per gallon water, apply as soil drench OR foliar spray. Repeat 2–3 times at 2-week intervals.
- ■ **Local source:** natural epsomite deposits found around mineral seeps in the Sulphur Springs Valley — identical to commercial Epsom salt.

8 · Seasonal Growing Calendar

Month-by-month tasks for 4,500 – 5,500 ft elevation

Month	Key Tasks	What's Happening
January	Start Rocoto and habanero seeds indoors. Set up heat mats. Apply elemental sulfur to beds if pH > 7.5 — needs weeks to work.	Rocoto seedlings germinating
February	Start all *C. annuum* varieties (jalapeño, cayenne, poblano, etc.) indoors. Continue soil amendment. Order drip irrigation supplies.	Pepper seedlings under grow lights
March	Pot up seedlings to 4" containers as they grow. Continue indoor growing. Last chance to apply sulfur before spring planting. Prep raised beds.	Indoor seedlings 4–6" tall
April	Begin hardening off late April. Check soil temp — wait for 60°F+. Transplant after last frost (late April at lower elevations). Install drip irrigation immediately.	Transplant to garden late April
May	Final transplants out. Mulch heavily. First fertiliser application. Watch for late frost — keep frost cloth ready until mid-May.	All varieties in ground; rapid growth
June	Peak heat and drought pre-monsoon. Monitor daily. First flower buds appear — switch to low-N fertiliser. Install shade cloth if temps consistently above 95°F.	Flowering begins; blossom drop risk
July	**MONSOON BEGINS.** Reduce drip irrigation immediately as rains arrive. Stake heavy-bearing plants. Watch for aphids in humidity. Rocoto begins flowering heavily — perfect conditions.	Fruit sizing rapidly; monsoon rains

Month	Key Tasks	What's Happening
August	Peak production for most *C. annuum* varieties. Harvest jalapeños, serranos, cayenne. Habaneros just beginning. Rocoto fruit approaching maturity.	Harvest most varieties; Rocoto ripening
September	★ **BEST MONTH.** Cool nights return. Pepper plants revive if stressed. Harvest red jalapeños (chipotles), fully-ripe poblanos (anchos), cayenne. Habanero peak harvest. Rocoto fully ripe.	Peak harvest for all varieties
October	Continue harvesting. First frost possible late October at higher elevations. Pot up Rocoto and overwintering plants before frost. Allow remaining peppers to fully ripen before cold.	Final harvest; frost watch begins
November	Frost ends outdoor growing at most elevations. Bring overwintering plants indoors. Dry and process remaining harvest. Apply compost to empty beds. Plant garlic.	End of season; preservation
December	Overwintered plants resting indoors under grow lights. Plan next season varieties. Order seeds. Test saved seeds from this season.	Indoor overwintering

9 · Canopy, Pruning & Support

Staking · pinching · shade cloth · increasing heat

Pepper plants at the Dragoons' productive elevation can become large and heavily laden. Providing the right support prevents broken branches and maximises fruit production. Strategic pruning and shade management make the difference between a struggling plant and a powerhouse producer.

Staking & Support

- ■ **Stake at transplant time** — not when plants are leaning. Late staking damages roots. Drive a 3-ft stake 6" from the plant.
- ■ **Cage large varieties** — poblano, bell, habanero, and Rocoto all benefit from tomato cages placed at transplant time.
- ■ **Tie loosely with soft cloth or Velcro ties** — wire or string cuts into stems. Retie as plant grows.
- ■ **Rocoto support:** these plants can reach 5–8 ft and become semi-vining — train along a trellis or fence for best management.

Pinching for Bushy Growth

- ■ **Pinch the growing tip at 6–8" height** — this forces branching and creates a bushy plant with many more fruiting sites than a single stem.
- ■ **Remove the first flower buds on transplants for 2–3 weeks** — redirects energy into root and canopy development, resulting in significantly higher total yield.
- ■ **Remove leaves below the first fork** — improves airflow, reduces fungal splash from monsoon rain, and focuses energy upward.

Shade Cloth Management

Situation	Solution
June pre-monsoon heat above 95°F	30% shade cloth over pepper beds. Reduces leaf temp 10–15°F and prevents blossom drop.
Rocoto in summer	40–50% shade on south and west sides. Rocoto needs shade during the hottest months.
Transplant establishment (May)	Light shade cloth first 2 weeks after transplanting reduces transplant shock from high UV.
Fall (Sept–Nov)	Remove all shade cloth. Full sun maximises ripening during the best growing season.

How to Make Peppers Hotter

The Dragoons' diurnal swings and sunny days already produce hotter chiles than most growing regions. These additional techniques maximise capsaicin:

- ■ **Reduce water in final 2–4 weeks before harvest** — mild stress increases capsaicin production significantly.
- ■ **Reduce nitrogen fertiliser after fruit set** — high-N dilutes capsaicin. Switch to potassium-focused feeding.
- ■ **Let fruit ripen fully on the plant** — red jalapeños are 2–3× hotter than green ones.
- ■ **Full sun maximises heat** — remove shade cloth once monsoon cools things down.
- ■ **Grow Chiltepin** — your native local variety is already among the hottest *C. annuum* peppers in existence.

10 · Pest & Disease Management

Common threats · organic controls · Dragoon-specific issues

Pest / Problem	Season	Signs	Organic Control
Aphids ★ Most common	Spring & Fall	Dense clusters on new growth; sticky honeydew; ants farming them	Strong water blast. Insecticidal soap spray. Neem oil. Plant basil nearby. Ladybugs (buy or attract naturally).
Spider mites	June (hot dry)	Tiny webbing under leaves; bronzed, stippled foliage	Increase humidity around plants (mist in morning). Neem oil or insecticidal soap. Avoid broad-spectrum pesticides.
Thrips	Spring–Summer	Silvery scars on leaves and fruit; distorted new growth; dropped flowers	Yellow sticky traps. Spinosad spray. Remove affected leaves. Blue sticky traps specifically target thrips.
Pepper weevil	Summer	Small holes in green fruit; fruit drops prematurely; internal larvae	Remove dropped fruit immediately. Neem oil spray on developing fruit. Pyrethrin as last resort.
Cutworms	Spring — transplant time	Seedlings cut off at soil level overnight	Cardboard collar around each stem 2” into soil. BT (*Bacillus thuringiensis*) drench at base. Diatomaceous earth.
Grasshoppers	June–Sept	Rapid defoliation — especially severe in drought years	*Nosema locustae* biological bait. Row covers. Chickens if nearby. Kaolin clay spray on leaves.

Pest / Problem	Season	Signs	Organic Control
Bacterial wilt (*Phytophthora*)	Post-monsoon rain	Sudden total wilting despite wet soil; dark basal crown	No cure — remove plant and soil immediately. Prevent: don't overwater; improve drainage; rotate solanaceae every 3 years.
Blossom end rot	Peak fruiting	Dark sunken spot on fruit bottom; watery, then dry and brown	See Chapter 7 — consistent watering + pH 6.0–7.0 prevents nearly all cases.
Sun-scald	June–August	White papery patches on fruit facing the sun	30% shade cloth in peak heat. Leave enough foliage to shade fruit. Orient rows north-south so plants shade each other.
Powdery mildew	Monsoon season	White powder on leaves — mainly in dense canopy	Improve airflow around plants. Dilute milk spray (1 part milk : 9 parts water). Potassium bicarbonate spray. Increase plant spacing.
Javelina	Year-round	Entire plants uprooted or eaten; strong musk smell	Low-voltage electric fence around bed perimeter. Welded wire cage around individual plants in first season.

11 · Harvest, Drying & Processing

When to pick · green vs. red · drying · powder · sauce

Most peppers can be harvested at two very different stages — green (unripe) or fully coloured (ripe) — and each gives a completely different flavour, heat level, and culinary use. The Dragoon fall season gives you excellent natural drying conditions for making chile powder and ristras.

Harvest Guide by Variety

Variety	Harvest Green?	Harvest Ripe?	Colour at Ripe	Dragoon Timing
Jalapeño	Yes — firm, dark green	Yes — red, sweeter and hotter	Red	Aug–Oct
Poblano	Yes ★ — classic use; dark green = poblano	Yes — dried red = ancho chile	Red/Dark Red	Aug–Oct
Serrano	Yes — green has brighter flavour	Yes — red/orange is hotter	Red or Orange	Aug–Oct
Cayenne	No — wait for red	Yes ★ — dry when fully red	Red	Sept–Oct
Anaheim / Hatch	Yes ★ — roast green	Yes — dried red = California chile	Red	Aug–Sept
Habanero	No — wait for full colour	Yes ★	Orange/Red/Chocolate	Sept–Oct
Chiltepin	No — wait for red	Yes ★	Red	Sept–Oct
Rocoto	No — wait for full colour	Yes ★	Red/Orange/Yellow	Sept–Oct (after monsoon cools)
Bell Pepper	Yes — green bell	Yes — red/yellow/orange; sweeter	Red/Yellow/Orange	Aug–Oct

Drying Methods for the Dragoon Climate

The Dragoon fall season — dry, warm days with low humidity — is perfect for traditional sun-drying and ristra-hanging. This is exactly the climate New Mexico chile growers prize.

Method	Best Varieties	How To	Time
Sun drying ★ Best in Dragoons	Cayenne, Chiltepin, Serrano, thin-walled chiles	Lay on wire racks in full sun. Bring in at night. Turn daily. Done when fully dry and brittle.	7–14 days
Ristra (hanging)	Hatch, Anaheim, Cayenne, Chiltepin	Thread through stems with needle and twine. Hang in shade with good airflow. Traditional Southwest method.	3–6 weeks
Oven drying	All varieties	Slice or leave whole. 150–170°F with door cracked. Check every 2 hours.	4–12 hours
Dehydrator	All varieties	125–135°F setting. Slice thick-walled varieties. Single layer on trays.	8–24 hours
Smoke drying (chipotle)	Jalapeño → Chipotle	Cold-smoke over local mesquite or oak 2–3 days. Dragoon area has excellent mesquite for this.	2–3 days

Local mesquite is abundant in the lower Dragoon foothills and Sulphur Springs Valley. Slow-smoking red jalapeños over mesquite wood for 2–3 days produces traditional chipotle with a distinct Dragoon terroir — an artisan product worth making.

12 · Overwintering Peppers

Indoor storage · containers · perennial management

Pepper plants are perennial in frost-free climates. At Dragoon elevation, frost ends outdoor growing in October–November, but bringing selected plants indoors for winter restarts them 6–8 weeks ahead of a new plant the following spring — and Rocoto plants are worth keeping alive for decades.

Which Varieties Are Worth Overwintering?

Variety	Worth Overwintering?	Why
Rocoto / Manzano (*C. pubescens*)	★★ Absolutely — treat as a permanent plant	Lives 15+ years. Gets more productive each year. Too slow-growing to replace annually.
Habanero / Scotch Bonnet	★★ Yes — strongly recommended	Takes 2+ years to reach peak production. Year-2 plants produce 3–4× more than seedlings.
Ghost Pepper / Extreme heat	★★ Yes — the only way to get reliable harvest	Needs 100–120 day season. Year-2 plants already at full size at bud break.
Chiltepin (native)	★ Yes, or let self-seed in protected microclimate	Perennial in Dragoons if sited against a south-facing wall or rock. Some plants survive mild winters in protected spots.
Jalapeño, Serrano, Cayenne	Optional — mainly for fun	Easy to grow from seed. Overwintering gives 4–6 week head start. Low priority.
Bell Pepper	Not worth it	Inexpensive seedlings. Poor indoor performer. Start fresh annually.

Overwintering Protocol

1. **Cut back hard before bringing in** — In early October, prune plants back to 6–8” stubs, leaving 3–4 main branches. This reduces transpiration and makes indoor plants manageable.
2. **Inspect and treat for pests** — Check carefully for mites, aphids, and scale before bringing inside. Treat with insecticidal soap spray and let dry before moving indoors.

3. **Pot up if needed** — Rocoto and habanero do best in 5-gallon containers minimum. Use well-draining mix: 60% potting soil + 40% perlite.
4. **Place under grow lights** — 16 hours of LED light per day. Plants are semi-dormant and need less than during summer.
5. **Water sparingly** — Every 7–10 days — just enough to prevent complete desiccation. No fertiliser until February.
6. **Resume feeding in February** — Start fish emulsion or compost tea in February. Plants will push new growth rapidly.
7. **Harden off and return outdoors in May** — Acclimatise for 1–2 weeks as you would a seedling. Year-2 plants will explode into production within weeks of going back out.

The Dragoon Mountains are ancestral pepper country. The wild Chiltepin has grown here for millennia, and the cool nights, monsoon rains, intense fall sunshine, and mineral-rich soils create conditions that produce peppers of exceptional flavour, heat, and character. From native Chiltepin growing wild in the washes to Andean Rocoto thriving in the cool evening air, this landscape was made for chile.